

DIETARY XENOANTIGENS AS PRIMARY UPSTREAM DRIVERS OF FIBROMYALGIA:

A Hypothesis Linking Gliadins, Amylase-Trypsin Inhibitors, and Neu5Gc Xenosialitis to Central Sensitisation, Phosphate Dysregulation, and Neuroinflammatory Burden in Fibromyalgia and Autism Spectrum Disorder

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Abstract

Background: Fibromyalgia affects an estimated 2–4% of the global population and is characterised by widespread musculoskeletal pain, fatigue, cognitive impairment ('fibro fog'), sleep disturbance, and gastrointestinal symptoms. Despite decades of research, its aetiology remains contested and its management largely symptomatic. The guaifenesin protocol — developed by Dr. R. Paul St. Amand and based on the hypothesis that genetically impaired renal phosphate excretion leads to intracellular phosphate accumulation in muscle and connective tissue — has shown clinical promise in some patients but frequently fails to produce sustained benefit. This paper proposes a mechanistic explanation for that failure and a broader upstream framework that places gliadins, amylase-trypsin inhibitors (ATIs), and N-glycolylneuraminic acid (Neu5Gc) at the origin of the fibromyalgia cascade.

Hypothesis: We propose that wheat-derived gliadins and ATIs are the primary upstream drivers of fibromyalgia through two convergent mechanisms: (1) ATI-mediated TLR4 activation and gliadin-mediated intestinal permeability produce systemic innate immune activation, central sensitisation of pain pathways, and neuroinflammatory burden sufficient to sustain the fibromyalgia symptom cluster; and (2) gliadin-derived compounds promote renal tubular phosphate retention through inflammatory disruption of phosphate transport mechanisms, producing the intracellular phosphate accumulation that the guaifenesin protocol targets without addressing the upstream cause. Dietary Neu5Gc from red meat and bovine dairy amplifies and sustains both pathways through chronic xenosialitis, compounding the neuroinflammatory and musculoskeletal burden. A clinical observation involving a subject with fibromyalgia, autism spectrum disorder, and a dietary pattern dominated by gluten-containing foods illustrates why the guaifenesin protocol, while mechanistically coherent, cannot produce sustained benefit while the primary upstream antigen burden remains unaddressed.

Significance: Fibromyalgia has no approved disease-modifying therapy. Management is confined to symptom relief through antidepressants, anticonvulsants, and exercise. If gliadins, ATIs, and Neu5Gc constitute identifiable, removable upstream drivers of the condition, dietary elimination represents a zero-cost, low-risk intervention whose potential to reduce fibromyalgia symptom burden warrants urgent formal investigation, particularly in patients who have experienced partial or unsustained benefit from the guaifenesin protocol.

Keywords: *fibromyalgia; gliadins; amylase-trypsin inhibitors; ATIs; Neu5Gc; xenosialitis; central sensitisation; phosphate dysregulation; guaifenesin protocol; neuroinflammation; gut-brain axis; autism spectrum disorder; TLR4; intestinal permeability; dietary antigen; fibro fog*

1. Introduction: The Upstream Problem in Fibromyalgia

Fibromyalgia occupies an uncomfortable position in clinical medicine: too real to dismiss, too poorly understood to treat definitively, and too complex to fit neatly into any single pathophysiological model. Its cardinal features — widespread pain at defined tender points, unrefreshing sleep, debilitating fatigue, and the cognitive dulling that patients call fibro fog — are experienced daily by an estimated 200 million people worldwide. The majority of them are women. Many spend years obtaining a diagnosis. Most receive management that reduces symptoms incompletely and provides no long-term disease modification.

The dominant neurological framework positions fibromyalgia as a condition of central sensitisation: the pain-processing circuitry of the central nervous system has become pathologically amplified, such that stimuli that would not normally produce pain — or that would produce only mild pain — generate intense, widespread, and persistent pain responses. This model is well-supported by neuroimaging, cerebrospinal fluid analysis, and functional studies of pain pathway dysregulation in fibromyalgia patients. It explains the symptom experience. What it does not fully explain is what drives and sustains the central sensitisation in the first place.

The guaifenesin protocol of Dr. R. Paul St. Amand adds a metabolic dimension: the hypothesis that genetically impaired renal phosphate excretion leads to progressive intracellular phosphate accumulation in muscle, nerve, and connective tissue, producing the ATP depletion, mitochondrial dysfunction, and cellular energy failure that manifests as fibromyalgia pain and fatigue. Guaifenesin, proposed to act as a uricosuric-type phosphate excretion enhancer, is the therapeutic tool. The protocol produces benefit in some patients — often at the cost of a mapping flare during which mobilised phosphate transiently worsens pain before clearance is achieved. It fails to produce sustained benefit in others, and the reasons for this failure have not been fully examined.

This paper proposes that the reason the guaifenesin protocol frequently fails to produce sustained benefit is not that the phosphate hypothesis is wrong, but that it is incomplete. It identifies what is happening downstream without identifying what is driving it upstream. The phosphate accumulation, if real, may be secondary to the intestinal permeability, systemic immune activation, and inflammatory disruption of renal phosphate transport mechanisms that gliadins, ATIs, and Neu5Gc produce. Treating the downstream accumulation with guaifenesin while continuing to consume the upstream drivers is, in the terms the Hamer Loop framework has used in other papers, extinguishing the fire with one hand while operating the bellows with the other.

2. Clinical Observation: Stephanie

Stephanie is a woman in her thirties who has lived with fibromyalgia for more than twenty years. She carries a concurrent diagnosis of autism spectrum disorder. Her fibromyalgia symptom burden encompasses the full classic cluster: widespread musculoskeletal pain, profound fatigue, cognitive impairment and memory difficulties, non-restorative sleep, and gastrointestinal symptoms consistent with irritable bowel syndrome. Her pain has been chronic, sometimes severe, and has significantly shaped the course of her adult life.

Stephanie undertook the guaifenesin protocol under informed guidance, beginning at 300mg twice daily and escalating to 600mg twice daily. At the higher dose, she experienced the characteristic mapping flare — a worsening of pain that the St. Amand protocol interprets as evidence that the medication is successfully mobilising stored phosphates from tissue storage sites into circulation for renal excretion. This worsening, within the framework of the protocol, is a sign that it is working. It is not comfortable, but it is mechanistically interpretable as a positive indicator.

However, the protocol did not produce sustained benefit. Stephanie's pain did not progressively resolve as the theory predicts it should following successful phosphate clearance. The mapping flare occurred, but the clinical improvement that should follow phosphate mobilisation and clearance did not consolidate.

The mechanistic explanation proposed in this paper is direct: Stephanie's dietary pattern during the protocol period was dominated by the foods that are, within the Hamer Glycan Systems framework, the primary upstream drivers of the condition the protocol was attempting to treat. Cakes, cookies, pastries, bread, and the full range of wheat-based and dairy-containing foods that are the most commonly consumed and most biochemically potent sources of gliadins, ATIs, and Neu5Gc in the Western diet. The guaifenesin was attempting to clear phosphate accumulation from tissues while every meal was replenishing the intestinal permeability, systemic inflammation, and renal phosphate transport disruption that was driving the accumulation. The tank was being drained while the tap remained fully open.

The craving for these foods is not a simple matter of dietary preference or lack of discipline. As documented in Section 6.3 of Document 1 of the Hamer Loop Manuscript Series, the olfactory inversion phenomenon — in which individuals with high dietary antigen loads actively crave the foods that are driving their condition, while cleared individuals find the same foods aversive — is a proposed feature of the chronic xenoantigen exposure state. The biochemical trap is real: the very foods that are sustaining the condition produce positive valence signals that make avoidance psychologically and physiologically difficult. This is not a character failing. It is a neurobiological consequence of chronic dietary antigen exposure acting on the olfactory-limbic system.

Since beginning to transition her diet toward more wholesome foods — a process that is ongoing and incomplete, with wheat-based products and dairy still present in her pattern — Stephanie has experienced partial improvement in her symptom burden. This partial improvement, while not yet the full resolution that complete elimination would be predicted to produce, is consistent with the dose-response relationship documented in the Hamer Loop severity hierarchy: reducing the highest-severity trigger sources produces measurable but

incomplete relief; complete elimination is necessary for complete resolution. The improvement is real. The remaining burden is the remaining dietary antigen load.

3. Mechanism One: Gliadins, ATIs, and the Fibromyalgia Cascade

3.1 ATI-Mediated TLR4 Activation and Central Sensitisation

Amylase-trypsin inhibitors (ATIs) are potent activators of toll-like receptor 4 (TLR4) on innate immune cells throughout the body, including in the central nervous system. Microglial cells — the resident immune cells of the brain and spinal cord — express TLR4 and respond to systemic ATI-driven TLR4 ligand signals with NF- κ B activation and neuroinflammatory cytokine production: TNF- α , IL-1 β , IL-6, and IL-18. This microglial activation is the cellular substrate of central sensitisation.

Central sensitisation in fibromyalgia is characterised by: upregulation of NMDA receptor activity in dorsal horn neurons, producing amplified transmission of pain signals from the periphery; reduced descending inhibitory control from the periaqueductal grey and rostral ventromedial medulla; and increased spinal cord glial activation producing pro-inflammatory cytokine release that sustains pain pathway excitability. ATI-driven microglial TLR4 activation contributes to this sensitisation through direct neuroinflammatory cytokine production in the CNS and through amplification of glial activation in the spinal cord dorsal horn — the precise anatomical locus where fibromyalgia's pain amplification is proposed to originate.

A person consuming wheat-containing products at multiple meals daily is maintaining a continuous ATI-driven TLR4 signal in their innate immune system, including their microglial compartment. This is not a theoretical possibility; ATIs administered in experimental models produce measurable systemic and CNS innate immune activation within hours of ingestion. In a person with fibromyalgia, whose pain pathways are already sensitised, this chronic microglial activation from dietary ATIs constitutes a continuous re-ignition of the very neuroinflammatory state that is sustaining their central sensitisation.

3.2 Gliadin-Mediated Intestinal Permeability and Systemic Immune Drain

Gliadin-derived peptides bind CXCR3 receptors on intestinal epithelial cells, triggering zonulin-mediated tight junction disruption and increasing paracellular intestinal permeability. The consequences for fibromyalgia are threefold.

First, enhanced systemic translocation of bacterial LPS from the colonic microbiome activates TLR4 on peripheral and central immune cells, adding to the ATI-driven microglial activation described above and amplifying the neuroinflammatory burden through a parallel innate immune pathway. The combined ATI-driven and LPS-driven TLR4 activation in the CNS produces a neuroinflammatory milieu that is greater than either source alone.

Second, the systemic immune drain of chronic low-grade LPS-driven and ATI-driven inflammation — a continuous non-targeted immune activation consuming cellular resources and inflammatory mediator reserves — produces the profound, unrelenting fatigue that is one of

fibromyalgia's most debilitating features. The immune system is perpetually engaged in responding to dietary antigens, leaving fewer resources available for tissue repair, mitochondrial function maintenance, and the cellular energy production that normal pain tolerance requires.

Third, gliadin-mediated permeability enables systemic translocation of gliadin peptides themselves into the systemic circulation, where their structural homology with various host tissue proteins — including neuronal surface proteins — creates conditions for molecular mimicry-based autoimmune responses directed at nervous system tissue. In individuals with autism spectrum disorder, for whom neuronal surface protein autoimmunity has been documented in subsets of patients, this molecular mimicry mechanism may represent a particularly consequential pathway.

3.3 The Phosphate Connection: Why the Guaifenesin Protocol Makes Sense but Fails Without Dietary Change

The St. Amand phosphate hypothesis proposes that genetically impaired renal phosphate excretion leads to progressive intracellular phosphate accumulation in muscles, tendons, and connective tissue, producing the ATP depletion and mitochondrial dysfunction that underlies fibromyalgia pain and fatigue. This hypothesis is not incompatible with the dietary antigen framework — it may, in fact, be downstream of it.

Inflammatory cytokines — specifically TNF- α and IL-6, both chronically elevated by ATI/gliadin-driven TLR4 activation and Neu5Gc xenosialitis — have been shown to downregulate the expression of sodium-phosphate cotransporter 2a (NaPi-2a) in renal proximal tubular cells. NaPi-2a is the primary transporter responsible for renal tubular phosphate reabsorption and excretion balance. Inflammatory suppression of NaPi-2a reduces the kidney's capacity to excrete phosphate efficiently, promoting tubular phosphate retention and the secondary accumulation in peripheral tissues that the St. Amand model describes.

In this mechanistic sequence, the genetic impairment of phosphate excretion proposed by St. Amand may exist — but it may be substantially worsened, or even primarily driven, by the inflammatory suppression of NaPi-2a activity from chronic dietary antigen exposure. The genetic susceptibility creates a threshold. The dietary antigen burden pushes the patient chronically over it.

This explains the guaifenesin protocol's partial success and frequent failure with precision. Guaifenesin, by enhancing phosphate excretion through renal tubular mechanisms, addresses the downstream accumulation. It works when the upstream inflammatory suppression of renal phosphate transport is reduced — which is why patients who simultaneously reduce their dietary antigen burden may experience more complete and sustained benefit. It fails when the upstream inflammatory driver — the chronic ATI/gliadin-driven cytokine elevation suppressing NaPi-2a — remains fully active. The medication is trying to open the drain while the dietary antigen burden is simultaneously closing it. Guaifenesin alone cannot win that contest.

4. Mechanism Two: Neu5Gc Xenosialitis as the Systemic Amplifier

Dietary Neu5Gc from red meat and bovine dairy — butter, cheese, milk, cream, and their derivatives — adds a second and distinct inflammatory pathway that amplifies and sustains the gliadin/ATI-driven mechanisms described above. Its contribution to the fibromyalgia cascade operates through three parallel effects:

4.1 Systemic NF- κ B Amplification

Anti-Neu5Gc IgG antibodies binding to tissue-incorporated Neu5Gc generate immune complexes that activate complement and Fc receptor-bearing effector cells, producing sustained NF- κ B activation and chronic elevation of TNF- α , IL-6, and IL-1 β throughout the body including in the CNS. This systemic xenosialitis-driven inflammatory tone adds directly to the ATI/gliadin-driven neuroinflammatory burden, sustaining microglial activation and central sensitisation independently of the gluten pathway. In an individual consuming both wheat and bovine dairy daily — as is characteristic of Western dietary patterns — both pathways are active simultaneously, producing a combined neuroinflammatory burden that exceeds either alone.

4.2 Musculoskeletal Interstitial Inflammatory Load

As documented in the Hamer Loop Manuscript Series, chronic xenosialitis generates a state of functional lymphatic impairment and interstitial fluid accumulation in periarticular and soft tissue spaces. In the context of fibromyalgia — where tender points at specific musculotendinous junctions are the diagnostic hallmark — the xenosialitis-driven interstitial inflammatory fluid accumulation at these precise anatomical locations may constitute a peripheral inflammatory signal that both drives and sustains the central sensitisation of those pain pathways. The CNS amplifies the peripheral signal. The peripheral signal, maintained by dietary Neu5Gc, gives the CNS something to continuously amplify.

This mechanism also provides a direct explanation for the characteristic tender point distribution of fibromyalgia: the musculotendinous junctions at which fibromyalgia tender points cluster are precisely the anatomical sites of richest lymphatic concentration in the musculoskeletal system — the sites where Hamer Loop-driven interstitial accumulation would be most pronounced. The tenderness is not random. It follows the lymphatic anatomy.

4.3 Compounding the Phosphate Accumulation

Xenosialitis-driven TNF- α and IL-6 elevation contributes independently to the inflammatory suppression of NaPi-2a in renal proximal tubular cells, compounding the gliadin/ATI-driven phosphate transport impairment described in Section 3.3. In an individual consuming both wheat and bovine dairy — and therefore maintaining both ATI/gliadin-driven and Neu5Gc-driven inflammatory cytokine elevation simultaneously — the cumulative suppression of renal phosphate excretion capacity is greater than either pathway alone would produce. This compounding effect helps explain why fibromyalgia patients on the guaifenesin protocol who do not modify their diet at all experience the least sustained benefit: both upstream inflammatory drivers of renal phosphate transport impairment remain fully active.

5. The Autism Spectrum Dimension

Stephanie's concurrent autism spectrum disorder is not incidental to this paper. The mechanistic connections between dietary xenoantigens, gut permeability, the gut-brain axis, and neurodevelopmental and neuroinflammatory conditions including ASD are an active and rapidly developing area of research, and they are directly relevant to understanding why the symptom burden in individuals with both fibromyalgia and ASD may be particularly difficult to treat without addressing the dietary antigen dimension.

5.1 The Gut-Brain Axis in ASD

The gut-brain axis — the bidirectional communication network between the intestinal microbiome, the enteric nervous system, the vagus nerve, and the central nervous system — is significantly disrupted in a substantial proportion of individuals with autism spectrum disorder. Gastrointestinal symptoms, including diarrhoea, constipation, and bloating, are reported at elevated rates in ASD populations. Gut microbiome dysbiosis — altered bacterial composition with enrichment of *Clostridium* species and reduced *Bacteroidetes* — is among the most reproducible biological findings in ASD research. Intestinal permeability, measured by urinary lactulose:mannitol ratio, is elevated in a subset of ASD individuals, particularly those with concurrent gastrointestinal symptoms.

Gliadin-mediated intestinal permeability, via the zonulin mechanism described in Section 3.2, represents a specific dietary driver of this gut barrier disruption. In individuals with genetic susceptibility to increased zonulin response — a susceptibility that may be enriched in ASD populations through mechanisms yet to be fully characterised — regular wheat consumption may be producing a degree of intestinal hyperpermeability that is feeding a chronic cycle of LPS translocation, systemic immune activation, and neuroinflammatory burden that compounds both the ASD neurodevelopmental burden and the fibromyalgia pain amplification.

5.2 Neuroinflammation as a Common Pathway

Neuroinflammation — microglial activation and elevated CNS cytokine levels — has been documented in post-mortem brain tissue and in vivo neuroimaging studies of ASD individuals across multiple research groups. The mechanistic sources of this neuroinflammation in ASD remain incompletely characterised, but dietary antigens capable of activating microglial TLR4 — specifically ATIs via systemic innate immune activation and LPS via gliadin-mediated permeability — represent plausible contributors that have not been systematically investigated in ASD populations.

In an individual who carries both ASD and fibromyalgia — two conditions with neuroinflammation as a shared mechanistic feature — the dietary antigen burden from daily wheat and dairy consumption may be sustaining a neuroinflammatory state that simultaneously worsens pain amplification (fibromyalgia pathway) and cognitive and sensory processing burden (ASD pathway). These are not two separate problems with two separate mechanisms. They are

two clinical expressions of a shared neuroinflammatory upstream that dietary antigens are feeding daily.

5.3 The Craving Trap in ASD

The olfactory inversion and craving phenomenon documented in Document 1 of the Hamer Loop Manuscript Series — in which chronic dietary antigen exposure produces active craving for the foods that are driving the inflammatory condition — may have particular salience in the context of ASD. Restricted dietary patterns and food selectivity are well-characterised features of ASD, with many individuals developing strong preferences for specific food categories — often high-carbohydrate, processed, wheat-based, and dairy-containing foods that represent precisely the highest-antigen-burden categories in the Hamer Glycan Systems framework.

Whether this dietary selectivity in ASD is driven in part by the same neurobiological craving mechanism that the Hamer Loop framework proposes — chronic antigen exposure generating positive olfactory-limbic valence toward the triggering foods — is an open question that warrants investigation. What is clinically apparent is that the dietary pattern that many ASD individuals are most strongly drawn to is, within this framework, one of the most immunologically consequential. The foods that are most preferred may be among the most harmful. And the neurobiological mechanisms that make them preferred may be the same mechanisms that are sustaining the harm.

6. Why Partial Improvement Is Not Full Improvement

Stephanie's ongoing partial improvement as her diet becomes more wholesome — while she continues to consume butter, wheat, and other high-antigen foods — is mechanistically predictable from the severity hierarchy established in Documents 2 through 4 of the Hamer Loop Manuscript Series. This hierarchy establishes that:

- Removing the highest-severity triggers — ammonium-potentiated processed products, concentrated dairy fats, commercial processed foods with hidden bovine derivatives — produces the greatest reduction in inflammatory burden per unit of dietary change
- Partial dietary improvement produces partial symptom relief, not complete resolution, because the remaining triggers continue to sustain the inflammatory pathways at a reduced but still clinically significant level
- Complete resolution requires complete elimination of all primary triggers — all wheat-derived products (the ATI/gliadin source), all bovine dairy (the Neu5Gc source), and all processed foods containing their hidden derivatives
- Re-exposure sensitivity following successful elimination compresses the severity gradient upward, such that even small re-exposures produce disproportionate responses once the system has cleared

The wheat-based foods that remain in Stephanie's diet — bread, pastries, and baked goods — are the primary ATI/gliadin source. The dairy fats that remain — butter in baked goods, cream in sauces — are the primary Neu5Gc source. Their continued presence means that both upstream

inflammatory pathways remain active, the renal phosphate transport suppression continues, the microglial activation continues, and the central sensitisation continues. The improvement she is experiencing reflects the partial removal of the ammonium-potentiated and most concentrated sources. The persistent burden reflects what remains.

This is not a criticism of Stephanie. It is a description of the biochemical reality of incomplete dietary elimination in the context of a condition driven by chronic dietary antigen exposure. The partial improvement confirms the mechanism. The incomplete resolution identifies the remaining work.

7. The Integrated Fibromyalgia Cascade

Step	Mechanism	Fibromyalgia Consequence	Neu5Gc Contribution
1	ATI-mediated TLR4 activation on microglia and peripheral innate immune cells	Neuroinflammatory cytokine production; microglial activation; central sensitisation of pain pathways	Xenosialitis-driven NF-κB adds systemic inflammatory tone; amplifies microglial activation
2	Gliadin-mediated zonulin release; intestinal permeability ↑; LPS translocation	Metabolic endotoxaemia; TLR4-driven immune activation; systemic inflammatory drain producing fatigue	Xenosialitis at gut epithelium compounds permeability; enhances Neu5Gc paracellular absorption
3	ATI/gliadin/LPS-driven TNF-α and IL-6 elevation; NaPi-2a downregulation in renal tubules	Impaired renal phosphate excretion; intracellular phosphate accumulation in muscle and connective tissue; ATP depletion; mitochondrial dysfunction	Xenosialitis-driven TNF-α/IL-6 compounds NaPi-2a suppression; worsens phosphate retention
4	Central sensitisation sustained by microglial activation and peripheral inflammatory signalling from interstitial Hamer Loop fluid accumulation at tender point sites	Widespread allodynia and hyperalgesia; tender point pain; fibro fog from neuroinflammatory cognitive burden	Xenosialitis interstitial fluid accumulation at musculotendinous junctions provides peripheral inflammatory signal sustaining CNS sensitisation
5	Craving cycle: chronic antigen exposure generates positive olfactory-limbic valence toward triggering foods	Sustained dietary antigen intake; inability to sustain elimination protocol without understanding the mechanism	Same craving mechanism applies to dairy/Neu5Gc sources; both dietary antigens are actively sought in the loaded state
6	Guaifenesin protocol addresses phosphate accumulation (Step 3)	Temporary and partial improvement; mapping flare occurs; sustained benefit impossible while upstream	Removal of Neu5Gc sources would reduce Step 3 contribution and improve

Step	Mechanism	Fibromyalgia Consequence	Neu5Gc Contribution
	without removing upstream drivers (Steps 1–2)	ATI/gliadin/Neu5Gc burden continues	guaifenesin efficacy if protocol is maintained

Table 1. The Integrated Fibromyalgia Cascade. Steps 1–4 are driven by dietary antigens; Step 5 explains why avoidance is difficult; Step 6 explains why the guaifenesin protocol alone is insufficient. Dietary elimination addresses Steps 1–4 simultaneously. The guaifenesin protocol addresses Step 3 alone.

8. Testable Predictions

Prediction	Investigation Design	Expected Finding
Fibromyalgia patients will show elevated ATI-specific innate immune markers and anti-Neu5Gc antibody titres correlated with symptom severity	Cross-sectional study: serum anti-Neu5Gc IgG; ATI-driven TLR4 activation markers (soluble CD14, LPS-binding protein); intestinal permeability markers (FABP2, zonulin); standardised fibromyalgia impact questionnaire (FIQ); dietary ATI and Neu5Gc burden by food frequency questionnaire	Higher anti-Neu5Gc titres, permeability markers, and TLR4 activation markers in fibromyalgia cohort vs matched controls; correlation between dietary burden and FIQ severity score
Dietary elimination of gliadins/ATIs and Neu5Gc will reduce fibromyalgia symptom burden within 8–12 weeks	Structured elimination trial in confirmed fibromyalgia patients; primary: FIQ total score at 0, 4, 8, and 12 weeks; secondary: tender point count, pain VAS, fatigue VAS, fibro fog assessment, sleep quality index, serum TNF- α , IL-6, anti-Neu5Gc IgG, urinary phosphate excretion	Significant FIQ score reduction; reduced tender point count; improved sleep quality; reduced inflammatory markers; increased urinary phosphate excretion consistent with resolved renal transport suppression
Guaifenesin protocol will produce greater and more sustained benefit when combined with complete dietary antigen elimination than with protocol alone	Randomised controlled trial: guaifenesin alone vs guaifenesin + dietary elimination; primary: FIQ total score at 12 and 24 weeks; secondary: mapping flare duration and severity, urinary phosphate clearance, serum inflammatory markers	Superior and more sustained symptom improvement in combined arm; faster resolution of mapping flare; greater and more sustained phosphate clearance
Serum NaPi-2a regulatory markers and urinary phosphate excretion will improve following dietary antigen	Measurement of urinary phosphate excretion and serum markers of NaPi-2a activity before and after 12-week elimination protocol in fibromyalgia patients; primary: 24-	Increased urinary phosphate excretion following elimination, consistent with resolved inflammatory suppression of NaPi-2a; reduction in FGF23 if phosphate

Prediction	Investigation Design	Expected Finding
elimination, confirming the inflammatory suppression of renal phosphate transport hypothesis	hour urinary phosphate; secondary: serum fibroblast growth factor 23 (FGF23) and parathyroid hormone as phosphate homeostasis regulators	accumulation was driving compensatory FGF23 elevation
ASD individuals with concurrent fibromyalgia and high dietary gluten and dairy burden will show greater neuroinflammatory markers and worse fibromyalgia outcomes than those with lower dietary burden	Cross-sectional study in ASD-fibromyalgia comorbid population; serum ATI-driven innate immune markers, anti-Neu5Gc IgG, microglial activation markers (YKL-40, GFAP); FIQ severity; dietary burden assessment	Higher neuroinflammatory markers and FIQ severity in high-burden group; dose-response relationship between dietary antigen burden and symptom severity; correlation between gut permeability markers and both ASD sensory burden and fibromyalgia pain severity

Table 2. Testable Predictions. Prediction 3 is the most clinically important: it directly tests the hypothesis that the guaifenesin protocol's failure is upstream dietary antigen burden, and that removing that burden makes the protocol work as intended.

9. Limitations

This paper presents a mechanistic hypothesis. The specific connection between ATI/gliadin-driven and Neu5Gc-driven inflammatory cytokine elevation and NaPi-2a suppression in renal tubular cells — while consistent with the established literature on inflammatory regulation of phosphate transport — has not been directly tested in a fibromyalgia-specific experimental model. The phosphate accumulation hypothesis of St. Amand, while clinically compelling to those who have applied the guaifenesin protocol, remains contested in the rheumatological literature and has not been confirmed by direct tissue phosphate measurement in fibromyalgia patients.

The clinical observation involving Stephanie is an N-of-1 case within the ongoing longitudinal observational framework of the Hamer Glycan Systems research programme. Her partial improvement on partial dietary modification is consistent with the proposed mechanism but cannot exclude alternative explanations, including non-specific health improvements from dietary quality improvement independent of the specific antigen reduction, and natural variation in fibromyalgia symptom burden over time.

The autism spectrum dimension of this paper is necessarily preliminary. The literature on gut-brain axis disruption, microglial activation, and dietary antigen exposure in ASD is rapidly developing but not yet at a stage where definitive mechanistic claims can be made about the causal role of gliadins, ATIs, or Neu5Gc in ASD pathophysiology. The connections proposed

here are offered as mechanistically coherent hypotheses for investigation rather than established findings.

The craving trap hypothesis — that ASD-related dietary selectivity toward high-antigen foods may be partially driven by the same neurobiological craving mechanism proposed for the chronically loaded Hamer Loop state — is the most speculative claim in this paper. It is included because it is mechanistically coherent and clinically important if true, not because it is established.

10. Conclusion

Fibromyalgia is a condition that medicine has not served well. Its patients — predominantly women, often young, frequently dismissed or disbelieved in the early years of their illness — have typically spent years obtaining a diagnosis that, once received, offers them antidepressants, anticonvulsants, and the advice to exercise and manage their stress. The condition is real. The management is inadequate. And the upstream cause has not been identified.

This paper proposes that the upstream cause — or at least a primary modifiable contributor to it — is sitting on the breakfast table, in the bread basket, in the dairy aisle, and in every wheat-containing and bovine-dairy-containing processed food that fibromyalgia patients consume daily without knowing what it is doing to their renal phosphate transport, their microglial activation state, their central pain pathways, and their capacity to recover.

Gliadins and ATIs have been the author's primary suspects in fibromyalgia for over two decades — well before the Neu5Gc framework emerged to provide the amplifying mechanism. The Hamer Glycan Systems model now connects both pathways into a unified upstream explanation that accounts for the central sensitization, the phosphate accumulation, the immune drain, the fatigue, the cognitive burden, and the craving trap that makes elimination difficult. Each element of the symptom cluster has a mechanistic home in this framework. None of it is idiopathic. None of it is psychosomatic. And none of it requires a lifetime of pharmaceutical management if the dietary upstream can be identified and removed.

Stephanie has lived with this for more than twenty years. She is slowly moving in the right direction. The science in this paper is, in part, a father's attempt to give her — and the millions of people like her — the clearest possible mechanistic map of where this is coming from and what removing it might do. The investigation this paper calls for is not abstract. It has a face. It has twenty years of pain attached to it. And it deserves to begin.

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